

“It Seems Like an Advertisement”: Attention, Interpretation, and Response to Mobile Antivirus Warnings

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Abstract. While the security community has achieved high technical precision in mobile malware detection, the effectiveness of these tools remains limited by the human-centric delivery of warnings. We conducted a task-based study with 23 participants using three commercial antivirus applications to investigate why accurate detection often fails to trigger successful mitigation. Guided by the Communication-Human Information Processing (C-HIP) framework, we analyzed user interactions across three dimensions: attention, interpretation, and response. Our results reveal a decoupling between detection capabilities and defense outcomes. First, detection modality largely dictates the success of the entire defense pipeline: automatic scanning achieved a 61–74% removal rate, whereas manual-scan workflows succeeded in only 22% of cases, primarily because the delivery mechanism failed to cross the threshold of user attention. Second, users systematically reinterpret legitimate security alerts as benign advertisements or routine notifications, driven by prior ad exposure and a lack of conceptual understanding of malware. Third, ambiguous action labels such as “**Resolve**” and “**Ignore**” foster conflicting mental models, leaving users uncertain about their security status even after following system prompts. These findings demonstrate that even near-perfect detection is of little use if the communication pipeline fails to bridge the gap to informed user action. We provide evidence-based recommendations to align mobile threat communication with human cognitive processes so that technical detection translates into actual device security.

Keywords: Mobile Security · Antivirus · Malware · Security Warnings.

1 Introduction

The rapid proliferation of smartphones has introduced new avenues for malware attacks, with mobile devices becoming prime targets for malicious software that

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disrupts functionality, extracts sensitive data, or gains unauthorized access [41]. Phishing—particularly through SMS-based smishing—remains one of the most effective attack vectors for mobile threats [34,37]. Attackers often impersonate trusted institutions, embedding malicious links in messages that prompt users to install trojanized applications. Unlike desktop environments, mobile devices impose inherent constraints that make it difficult for users to distinguish between legitimate and malicious content. For example, smaller screens and touch-based interaction limit users’ ability to inspect suspicious URLs or link details, while widely trusted communication channels like SMS are particularly susceptible to exploitation [46].

Antivirus software is a core component of mobile security, designed to detect and mitigate these threats [44]. However, the effectiveness of technical detection may be fundamentally limited by the human-centric delivery of security warnings. Prior work has shown that mobile users frequently ignore or misinterpret warnings due to ambiguous language or habituation [3,24]. In regions like South Korea, these challenges are compounded by regulatory environments where mandatory security tools often operate as background plugins, potentially fostering passive engagement [25].

Despite the high precision of modern detection algorithms, a gap remains at the final stage of the defense process, where technical alerts must be translated into informed user actions. We therefore investigate how users interpret and respond to security warnings in realistic mobile threat scenarios, using the Communication-Human Information Processing (C-HIP) model [43] to examine whether detection capabilities and actual defense outcomes are decoupled. We structure our analysis around three functional dimensions: attention, interpretation, and response. To examine these interactions empirically, we conducted a task-based study with 23 smartphone users, addressing the following research questions:

- **RQ1 (Attention):** How does the warning delivery method affect users’ initial recognition of mobile malware threats?
- **RQ2 (Interpretation):** How do users interpret the nature and severity of mobile antivirus warnings, and what factors shape their understanding?
- **RQ3 (Response):** How do action labels and interface design elements influence users’ threat mitigation decisions?

By observing user responses in realistic phishing scenarios, we identify instances where technical detection fails to translate into effective mitigation, revealing that current mobile antivirus designs can inadvertently hinder timely user intervention. In terms of *attention*, automatic detection workflows were far more effective (61–74% removal) than manual-scan paradigms (22%), which often failed to cross the threshold of user awareness. In terms of *interpretation*, participants frequently misattributed legitimate alerts to benign sources such as advertisements or routine notifications. In terms of *response*, ambiguous labels such as “Resolve” and “Ignore” fostered conflicting mental models, leaving users uncertain about their security status even after following system prompts.

Our primary contributions include: (1) empirical evidence of a gap between technical detection capabilities and successful threat mitigation in real-world mobile contexts, (2) the finding that users systematically reinterpret legitimate antivirus warnings as advertisements or routine system notifications, a misattribution pattern distinct from the habituation effects documented in prior warning research, and (3) a within-subjects comparison showing that automated workflows outperform manual scanning, with up to 52 percentage points higher success in malware removal. Based on these findings, we provide recommendations to align technical security tools with human cognitive processes.

2 Related Work

2.1 User Perception and Mobile Security Behavior

Mobile users often demonstrate limited risk awareness and make suboptimal security decisions due to insufficient technical knowledge or overconfidence [10,42]. While theoretical frameworks such as Protection Motivation Theory (PMT) have been used to analyze threat appraisals [36,48], most prior studies rely on surveys, which may not fully capture real-time decision-making processes under active threat [18,30]. This gap is critical on mobile platforms where malware spreads through distinct, context-dependent vectors like SMS phishing and permission abuse [33]. These behavioral challenges are especially pronounced in South Korea; mandatory security software for financial services often relegates these tools to background “plugins,” fostering passive engagement and conceptual confusion between antivirus and general optimization tools [25]. While previous work has explored adoption factors and general perceptions, how users cognitively process and respond to warnings in active-threat scenarios remains understudied. Our task-based study addresses this gap directly.

2.2 Mental Models and Risk Communication in Mobile Security

Effective risk communication in computer security requires aligning with non-expert users’ mental models rather than expert-defined representations [7]. These mental models, the internal representations users hold of system behavior, often diverge from technical reality [13,31]. In mobile contexts, this mismatch is exacerbated by hidden system operations and abstracted security features [47,22]. Consequently, users with weak mental models frequently act on inaccurate assumptions, increasing their vulnerability to deceptive applications and misleading notifications that mimic legitimate behavior [19,14].

Users interpret security mechanisms through personal assumptions rather than technical properties. They misread HTTPS as an absolute guarantee of safety [28] and overestimate the anonymity of private browsing [1]. Evaluating the effectiveness of mobile antivirus warnings therefore requires examining how users interpret these alerts through their internal mental models, rather than relying on interface design analysis alone.

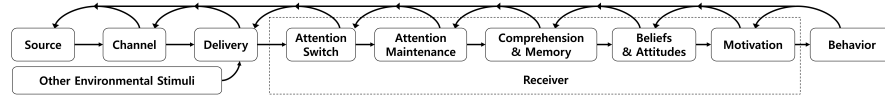


Fig. 1: Communication-Human Information Processing (C-HIP) model, adapted from [43].

The C-HIP model explains how warning information is processed from its source to the user’s final behavior [43] (see Figure 1). A source initiates the warning, which is transmitted via a channel (*e.g.*, visual or auditory signals) and delivered to the user. At the receiver level, users must first attend to the warning, maintain focus long enough to extract relevant information, and correctly comprehend its meaning. This understanding is then filtered through existing beliefs and attitudes, which shape the motivation to act and ultimately the behavioral response. Because a failure at any stage can disrupt information flow and lead to unsafe outcomes, the C-HIP model provides a framework for diagnosing breakdowns in warning effectiveness.

Building on this model, we map its processing stages into three functional dimensions, *attention*, *interpretation*, and *response*, to structure our analysis. *Attention* captures whether users notice the alerts; *interpretation* combines comprehension and beliefs to reflect how users understand and evaluate the threats; and *response* integrates motivation and behavior to capture how users decide and act. This mapping lets us analyze interaction failures at a higher conceptual level while remaining grounded in the underlying C-HIP processing structure.

2.3 Security Warning Design and Usability Barriers

Mobile users frequently ignore or misinterpret warnings due to habituation, misplaced trust, or device-specific interface constraints [4,3,15]. Field studies indicate a high rate of “click-through” behavior, particularly as users become desensitized to frequent alerts [20,35]. A primary usability barrier is the reliance on technical jargon; ambiguous terms lead to misconceptions about threat severity and the consequences of user actions [21,24]. On small screens, simplified UI elements often omit critical security metadata, further hindering appropriate responses [45,50]. To mitigate these issues, researchers have proposed design metaphors and affective cues to link technical risks to intuitive experiences [12,49,40,29]. However, many commercial mobile antivirus designs continue to obscure critical information [38]. Our study examines how these documented design flaws manifest in real-world smishing scenarios.

3 Methodology

We conducted a task-based, within-subjects study with 23 participants to examine how users perceive, interpret, and respond to mobile antivirus warnings.



Fig. 2: Comparison of the official IBK Bank mobile website (left) and a phishing replica (right). While visually identical, the top-left button labels differ (highlighted in red): “Banking Login” (뱅킹로그인) on the official site versus “Mobile Application” (모바일신청) on the phishing site.

Participants interacted with simulated malware on study-provided Google Pixel 3a smartphones preloaded with three commercial antivirus applications, encountering identical threat scenarios across all conditions. This design enabled a controlled comparison of interface effects while capturing natural user behavior under institutional review board approval and the ethical safeguards detailed below.

3.1 User Task Design

Phishing Scenario and Artifact Design. Our study simulated realistic malware threats using finance-related SMS phishing attacks, which are prevalent in real-world scams and significantly impact users [34,37]. The scenario was based on documented real-world attacks in which adversaries impersonate legitimate banks, lure users with loan-related messages, and induce them to install APK-based malware [17]. This attack pattern reflects recent trends on the Android platform, where banking Trojans are increasingly distributed via APK-based malware [26]. To faithfully reproduce a realistic attack context, we used a Trojanized APK impersonating a well-known domestic bank in South Korea.

We employed a single fixed scenario, using the same bank brand, smartphone model, and malware, so that observed differences in user behavior could be attributed to variations in antivirus warning interface design rather than to the attack context. To maximize ecological validity, we developed high-fidelity phishing artifacts, including SMS messages and a website impersonating the

target bank. The SMS informed recipients of their eligibility for a government-supported special loan, instructing them to click a link and select the “**Mobile Application**” button on the subsequent page to install the required app.

As shown in Figure 2, the phishing website closely mimicked the official bank interface, with subtle differences in button labeling (“**Banking Login**” vs. “**Mobile Application**”) reflecting common phishing tactics. While the official banking website either opens the legitimate bank app or redirects users to the Google Play Store, the “**Mobile Application**” button on the phishing site initiated the download of a malicious APK hosted on our lab server, thereby triggering the preinstalled antivirus software.

Antivirus Software Selection and Interface Profiles. To capture a broader range of user behaviors and identify interface features that enhance or hinder warning effectiveness, the study incorporated three widely used antivirus applications in South Korea: Avast Antivirus & Security (Avast, Version 23.3.2) [8], V3 Mobile Security (V3, Version 3.5.3.1) [2], and Kaspersky: VPN & Antivirus (Kaspersky, Version 11.91.4.9037) [27]. These applications were chosen to represent diverse user interfaces, detection methods, and warning designs, and were tested using only their free-tier features without any paid options.

When users attempt to download files via web links, Avast and V3 automatically initiate malware scans and display immediate warnings if threats are detected. Kaspersky, in contrast, does not support automatic scanning; users must manually trigger a scan through its on-demand interface. This difference in detection modality, rather than any difference in detection accuracy, is what our study exploits: all three applications flagged the same trojanized APK. Figures 3 to 5 compare the warning flows across the three applications.

Avast performs automatic scanning and immediately presents a bold “**Malware detected!**” warning containing malware details and two clearly labeled action buttons: “**Ignore**” and “**Resolve**.” This binary choice appears at every step of the warning flow, from the initial notification to the in-app threat list, giving the risk-accepting option the same prominence as the secure one throughout.

V3 also initiates automatic scanning, but its warning process is distributed across multiple screens. It first issues a “**Security threat detected**” notification, followed by a summary screen identifying the app as a Trojan and a detail screen listing the malware name, type, status, and file path. Both screens offer a single “**Delete**” button, providing no equivalent to Avast’s “**Ignore**” option and limiting user discretion.

Kaspersky, by contrast, requires the user to start the scan in its free-tier version: it is triggered through the “**Scan the new app IBK 기업**” notification element located in the notification drawer. This drawer serves a dual purpose: it contains not only scan notifications but also general system recommendations that prompt users to adopt Kaspersky’s recommended configurations. These settings notifications, indicated by messages such as “**2 issues**” in red text, encourage users to optimize their security setup by actions such as adding the Kaspersky app to the battery-saving exclusion list. Upon detecting a threat, the

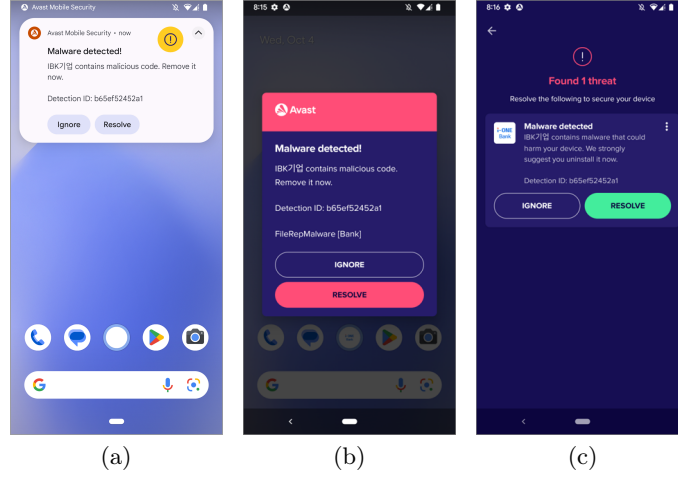


Fig. 3: Warning UI of Avast Antivirus & Security. Avast scans automatically on download: (a) a threat notification names the flagged app, (b) the warning screen adds a detection ID and malware label, and (c) the in-app threat list repeats the same information. All three steps pair **Ignore** with **Resolve**, giving the risk-accepting option equal visual weight.

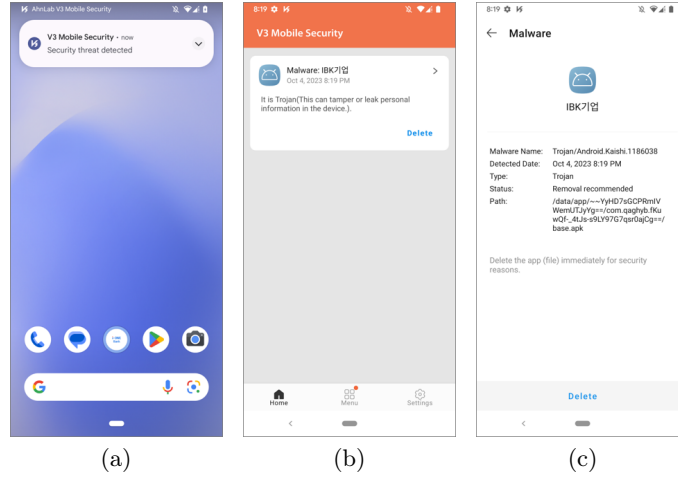


Fig. 4: Warning UI of V3 Mobile Security. V3 also scans automatically but spreads the warning across screens: (a) the notification reads **Security threat detected**, (b) a summary screen identifies the app as a Trojan, and (c) a details screen lists the malware name, type, and file path. Both (b) and (c) offer **Delete** as the only action, with no risk-accepting alternative.

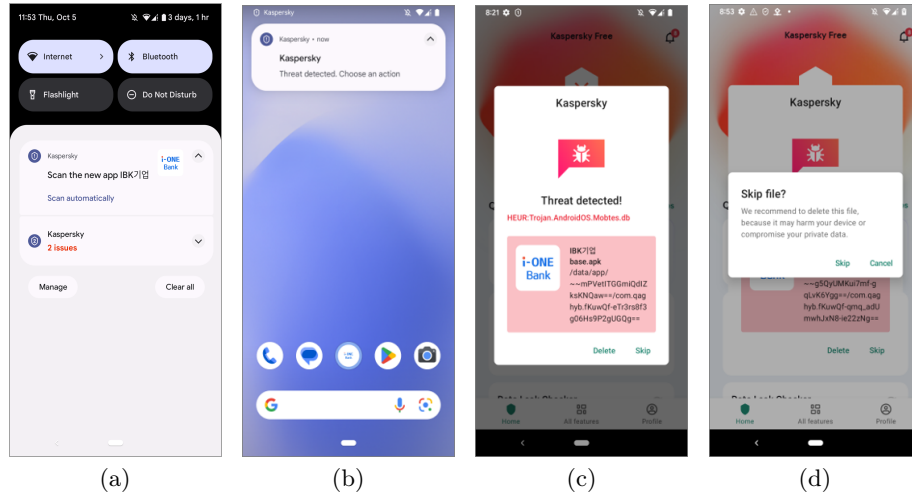


Fig. 5: Warning UI of Kaspersky VPN & Antivirus. Kaspersky does not scan automatically: (a) the scan prompt sits in the notification drawer next to a configuration recommendation marked **2 issues** in red, (b) a threat notification appears only after the user runs the scan, (c) the warning offers **Delete** and **Skip**, and (d) selecting **Skip** triggers a confirmation dialog.

system displays a prominent “**Threat detected!**” warning with red iconography, accompanied by “**Delete**” and “**Skip**” action buttons. If the user selects “**Skip**,” a follow-up confirmation dialog prompts them to confirm or revise their decision before proceeding.

All three applications require an explicit user decision to mitigate a detected threat, but they differ in presentation, information disclosure, and interaction complexity. These differences let us examine how variations in threat communication shape user responses to the same underlying detection.

3.2 Study Protocol

Our study followed a three-stage protocol as illustrated in Figure 6. Each session lasted approximately two hours and was conducted individually with participants using study-provided Google Pixel 3a smartphones running Android 12 to minimize concerns regarding malware installation on personal devices.

Stage 1: Orientation and Pre-task Interview. Participants first provided informed consent and were briefed that the study investigated their experiences with mobile antivirus software. We assessed their prior experience with mobile security applications through preliminary questions. Participants then received a device orientation to ensure comfortable use of the study phone, familiarizing them with basic operations such as navigation, the home button, and viewing active processes.

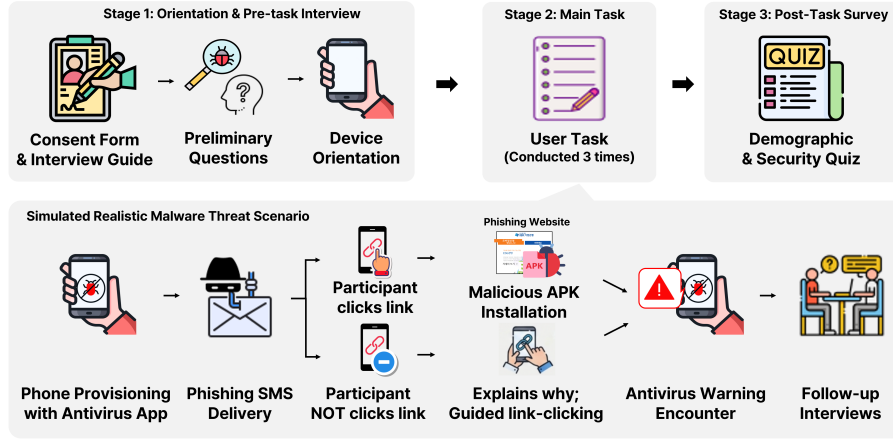


Fig. 6: Overview of our study protocol: (1) orientation and pre-task interview, (2) main task with three antivirus applications, and (3) post-task survey.

Stage 2: Main Task. We simulated realistic malware threats by sending mock phishing SMS messages containing links to malicious APK files. Participants were instructed to respond naturally, as they would in their daily lives. The task followed two paths: participants who clicked the phishing link navigated through a malicious website, downloaded the APK, and encountered antivirus warnings during installation. Those who initially avoided clicking were asked to explain their reasoning and were subsequently guided through the link-clicking process to ensure that all participants experienced the antivirus warning interfaces, maintaining consistency in data collection.

We observed participant responses throughout the session, focusing on their interaction with warning messages, decision-making processes, and any difficulties experienced. Semi-structured follow-up interviews immediately explored their understanding of warnings, perceived risks, and response reasoning. During these interviews, we showed participants the specific warning screens they had viewed to ensure accurate recall and facilitate detailed discussion. This task was repeated three times per participant, once for each antivirus application (*i.e.*, Avast, V3, and Kaspersky). The order of applications was counterbalanced using a Balanced Latin square design [11], ensuring that each application appeared equally often in each position across participants to mitigate learning and order effects. Participants were instructed to approach each session independently and respond as if encountering the warnings for the first time.

Stage 3: Post-Task Survey. Finally, participants completed demographic questionnaires and a security knowledge quiz adapted from prior work [32] to assess how background knowledge influenced threat responses. The complete questionnaire is provided in Appendix.

Table 1: Summary of participant characteristics ($N = 23$). Security Score is the number of correct answers on the three-item security knowledge quiz.

Category	Characteristic	n	%
Gender	Male / Female	11 / 12	47.8 / 52.2
Age	19–38 / 39–58 / 59+	8 / 11 / 4	34.8 / 47.8 / 17.4
Education	Below bachelor’s / Bachelor’s / Graduate	8 / 12 / 3	34.8 / 52.2 / 13.0
CS Background	Yes / No	2 / 21	8.7 / 91.3
Security Score	Low (0–1) / High (2–3)	10 / 13	43.5 / 56.5
Antivirus Usage	Adopted / Discontinued / Never	11 / 8 / 4	47.8 / 34.8 / 17.4

3.3 Participants

Participants were recruited via university announcements and a local online marketplace in South Korea. To mitigate response bias, the study was framed as an investigation into antivirus effectiveness, with phishing tasks undisclosed during recruitment. Our recruitment materials targeted experienced Android users without formal security education to capture the behaviors of casual users with limited security backgrounds. We assessed participants’ security background through the background questions and knowledge quiz in Stage 3.

Following established qualitative security studies [1,5,22], we targeted 24 participants. This size exceeds the minimum requirement of 18 for a balanced Latin square design [11], reducing order and carryover effects across the three antivirus interfaces. One participant was excluded due to insufficient Android experience, resulting in a final sample of $N = 23$. Each participant received 50,000 KRW (approx. 38 USD) for the two-hour session.

As summarized in Table 1, participants (11 male, 12 female) ranged from 24 to 68 years old. The majority (91.3%) lacked professional IT or computer science experience, with only two exceptions (P5, an iOS developer; P17, a CS major). The security quiz indicated only moderate familiarity overall ($M = 1.78$, $SD = 0.83$ out of 3), and 47.8% were active antivirus users.

3.4 Interview Data Analysis

We analyzed the interview data using an inductive, iterative qualitative coding approach [16]. The first author independently conducted initial coding to develop a preliminary codebook grounded in the dataset. Subsequently, the second author independently coded the data and provided feedback, resulting in refinements to the code definitions and the organizational hierarchy. Through multiple rounds of discussion, the authors reached a consensus on a final set of codes. All coding discrepancies were resolved through deliberation, and multiple codes were assigned when necessary to fully capture the complexity and nuances of the participants’ accounts.

3.5 Ethical Considerations

The study was approved by the Institutional Review Board (IRB) of Sungkyunkwan University and conducted in accordance with ethical standards for human-

subject research. To protect participant privacy, all sessions were conducted in a lab setting using research-provided smartphones. No personal devices or accounts were accessed during the study, and we did not collect any personally identifiable information, such as names, phone numbers, or login credentials. The only recorded data were screen activity and audio, captured with participant consent.

After each session, we debriefed participants and explained that all bank-related SMS messages, websites, and apps used during the tasks were created by the researchers for simulation purposes. We clarified that these scenarios did not represent actual institutions or the real behavior of the tested antivirus software. This setup enabled realistic evaluation of user responses while maintaining full transparency and minimizing participant risk.

4 Study Results

We first analyze users’ initial reactions to phishing attempts, and then investigate the root causes of interaction failures in antivirus software across three dimensions: attention, interpretation, and response, as summarized in Table 2.

4.1 User Responses to Phishing SMS

We begin by examining how users initially responded to simulated phishing messages that led to malicious APK downloads. Of the 23 participants, 15 successfully recognized the phishing attempt and refrained from clicking the link, often citing prior experience or general caution regarding SMS-based threats. However, 8 participants demonstrated behaviors that could have led to security compromises. Six participants expressed conditional willingness to click, particularly if the offer aligned with their personal needs, such as actively seeking loans (P10), being existing customers of the referenced bank (P21), or finding the message convincing after an online search (P15). As P16 noted, *“Even if the text looks weird, I’d at least check the loan, since I don’t think there’s much risk in just clicking the link.”*

To our surprise, two participants (P14, P18) clicked the link without hesitation. P14 did so based on trust in the bank, while P18 believed the message was related to a legitimate government loan program. These findings illustrate how contextual cues and perceived institutional legitimacy can override security awareness, exposing vulnerabilities in mobile phishing contexts.

4.2 Interaction Failures: Attention

The results of the main task observations indicate that the method by which antivirus programs deliver warnings is a critical factor in determining users’ initial recognition of security threats. Users’ recognition was strongly influenced by automated detection and the visual salience of alerts, both of which are essential for effective threat mitigation.

Table 2: Categories of failure outcomes and causes in user interaction with mobile antivirus warnings.

Dimensions	Outcome	Failure Causes
Attention	Threat Exposure Failure	– Missing auto-scan & Low accessibility: Hidden prompts and manual scan requirements preclude initial discovery.
	Overshadowed	– Visual competition & Distraction: Routine notifications compete visually with security alerts, obscuring them and leading to misinterpretation.
Interpretation	Threat/Warning Misunderstanding	– Lack of malware knowledge: Users are unfamiliar with malware and confuse it with emails, app names, or software bugs. – Visual mimicry & Prior ad-exposure: Prior exposure to ads undermines authenticity, leading users to perceive warnings as deceptive advertisements. – Contextual over-trust: Familiar branding leads users to misinterpret threats as normal technical issues.
	Underestimation of Risk	– Insufficient risk description: Vague explanations cause users to trivialize threats. – Perception of antivirus oversensitivity: Alerts are perceived as false positives. – Linguistic affect heuristics: Users judge severity based on wording (e.g., “security threat” vs. “malware”).
	Execution Failure	– Ambiguity of labels: Labels like “Resolve” or “Ignore” fail to convey actual effects. – Lack of behavioral guidelines: Absence of guided steps leads to confusion or false assumptions of auto-resolution. – Undisclosed system failure: System failures are not communicated, leaving users unaware that threats persist.
Response	Avoidance/Distortion	– Fear of operation: Users avoid interaction due to fear of worsening infection or damage. – Insufficient motivation: Absence of clear evidence leads to delayed decision-making. – Cognitive fatigue: Users follow system prompts without evaluation, either due to uncertainty or cognitive fatigue.
	Confusion/Uncertainty	– Complex Technical Info: Technical jargon overwhelms users, causing confusion or avoidance. – Lack of Feedback: Absence of confirmation leaves users uncertain and anxious.

Avast and V3 automatically initiate scans upon file download and notify users without requiring manual intervention. This proactive approach improved malware removal rates: 14 out of 23 participants (61%) using Avast and 17 out of 23 (74%) using V3 successfully eliminated the threat by selecting actions such as “Resolve” or “Delete.” V3 achieved a higher removal rate than Avast despite similar detection mechanisms, which participants attributed to interface simplicity; V3 presents a single “Delete” button, whereas Avast’s binary choice (“Ignore” and “Resolve”) introduced decision friction and hesitation. This pat-

tern echoes findings from Felt *et al.* [21], where users were more likely to proceed safely when only one clear, secure option was presented.

Kaspersky, in contrast, does not perform automatic scans and instead requires users to manually locate and initiate the scan function (see Figure 5). This missing auto-scan, combined with the low accessibility of the scan prompt hidden within the notification drawer, led to a threat exposure failure, in which users failed to recognize the presence of a threat. A majority of participants (74%, 17/23) never opened the notification panel, and among the six who did, only two correctly recognized and acted on the warning, yielding a 91% (21/23) failure rate in locating and using the scan function. P2 questioned this burden, stating, “*I don’t understand why it’s asking me to scan when it’s their job.*” P19 similarly remarked, “*I have to pull down the top bar to see the scan notification. If it’s preventive antivirus software, it should be more visible.*”

Consequently, only 5 out of 23 participants successfully removed the malware, roughly a third of the rate achieved under automated detection. Although our sample size is optimized for qualitative insights, a within-subjects comparison across the three applications supports this contrast (Cochran’s $Q(2) = 16.71$, $p = 0.0002$), and exact McNemar tests with Holm correction confirm that both automated applications outperformed manual scanning (V3 vs. Kaspersky: $p = 0.0014$; Avast vs. Kaspersky: $p = 0.0078$). Notably, only two of the five successful cases in the Kaspersky condition were triggered by the scan prompt itself; the remaining three participants independently identified the threat during installation (*e.g.*, by noticing excessive permission requests, abnormal URLs, or unknown sources). P11 stated, “*I expected a notification about the downloaded app, but there was none, so I checked manually,*” while P8 reported, “*I just wanted to check whether the IBK Bank app was fully installed,*” indicating that the scan was performed incidentally. Kaspersky’s effectiveness therefore relied on user initiative and serendipity rather than on the system surfacing the threat.

Kaspersky’s interface also led to critical attribution errors, as security alerts competed with routine notifications for visual attention. Many participants were distracted by prominent red-text settings notifications (*e.g.*, “2 issues”), which overshadowed the actual threat. P5 explained, “*I ignored the security alert because I thought it was just another routine issue like the ‘2 issues’ notification that wasn’t urgent.*” More problematically, similar visual treatments led P13 to a dangerous misunderstanding: “*I saw it detected two malware files, but the system only removed one. I thought I needed to find and delete the second one myself, so I reinstalled the suspicious app to locate it... but then I couldn’t figure out how to remove it properly.*” P13’s decision to reinstall suspected malware illustrates how interface confusion directly compromises device security.

Answer to RQ1: The detection delivery method largely dictates awareness and mitigation success: automated scanning achieved 61–74% malware removal, whereas manual workflows succeeded in only 22% of cases, owing to hidden scan prompts, low notification visibility, and visual competition with routine alerts. Among automated systems, the single-action interface

(V3, 74%) showed a higher removal rate than the binary-choice design (Avast, 61%), which participants attributed to reduced decision friction. Effective systems must default to automatic detection and establish a visual hierarchy that reflects threat urgency.

4.3 Interaction Failures: Interpretation

We explored how participants interpreted mobile antivirus warnings and understood the nature of the threats being presented. Our findings show that users do not simply fail to comprehend mobile antivirus warnings; rather, they systematically reinterpret them through prior experiences, contextual assumptions, and subjective judgments of severity.

While most participants recognized that a warning signaled a potential risk, 18 individuals had never encountered the term “malware,” leading to widespread conceptual confusion. For example, P7 associated malware with email, asking, “*Does this mean my email is infected?*” while P3 believed that “malware” referred to the name of the flagged application. Others described the threat vaguely as “something wrong” (P6, P8, P19) or conflated it with ordinary software flaws. As P16 explained, “*Where can we find a 100% clean app? I think that during app development, it is common for 1–2 instances of malware to be present in every app.*”

This confusion extended to the interpretation of the warnings themselves. Visual mimicry and prior ad exposure led users to dismiss legitimate alerts as advertisements. For instance, P4 stated, “*It seems like an advertisement... if I click it as suggested, I think an ad will pop up saying ‘You’re currently using the free version.’*” Conversely, some participants reframed the warning as a technical or compatibility issue due to contextual over-trust. P14, for example, described the alert as “*a conflict between the bank’s security level and the phone,*” effectively reinterpreting a security threat as a benign system conflict.

Even when participants acknowledged a potential risk, they frequently underestimated its severity. When warnings lacked explicit consequences, users questioned whether the threat involved serious harm or a minor inconvenience, highlighting the negative impact of insufficient risk descriptions. As P16 noted, “*It says malware, but there’s no explanation of what it actually does... it could steal my data, or just drain the battery,*” indicating that the threat was perceived as a performance issue rather than a security compromise. Furthermore, hedging expressions such as “**may harm**” reduced the perceived urgency: “*If it says ‘may harm,’ doesn’t that mean there might be no threat at all?*” (P4).

Participants also attributed warnings to antivirus oversensitivity, treating them as system overreactions rather than genuine threats. As P16 explained, “*Antivirus apps are usually overly sensitive ... it’s just being cautious because it’s a banking app.*” Finally, linguistic affect heuristics heavily influenced severity judgments. Participants distinguished between “security threat” and “malware,” consistently perceiving the former as less severe. For example, P21 dismissed the warning by stating, “*It’s just a security threat, not malware.*” Ultimately, these

reinterpretations directly dictated user behavior, leading to warning dismissal, risk acceptance, or inaction based on flawed subjective judgments.

Answer to RQ2: Users interpret antivirus warnings through subjective mental models rather than technical accuracy. Limited malware knowledge, prior ad exposure, and contextual assumptions lead users to misinterpret warnings and underestimate their severity. These reinterpretations directly drive behavior, resulting in warning dismissal, risk acceptance, or inaction.

4.4 Interaction Failures: Response

Execution Failure. We observed instances where users’ behavioral intentions failed to translate into successful threat resolution due to interaction design constraints. These failures were primarily driven by ambiguous action labels, a lack of behavioral guidance, and undisclosed system errors.

Conceptual misunderstandings directly influenced how participants interpreted and responded to action prompts. Even seemingly straightforward options like “Resolve” and “Delete” were frequently misinterpreted. For example, P6 believed “Resolve” implied a partial fix—removing malicious code while leaving the application intact—whereas P18 viewed the term as a security measure provided by the bank rather than the antivirus application. Ambiguous labels such as “Ignore” and “Skip” also proved problematic; P13 assumed “Ignore” functioned as a mechanism to block malware execution. These examples demonstrate how unclear terminology can lead users to take actions based on a false sense of security.

When systems detect threats without providing explicit behavioral guidance, users experience confusion and uncertainty about how to proceed. In V3, P4 did not realize that tapping the “Security threat detected” notification would initiate remediation, instead attempting to remove the malware manually and ultimately failing. As P4 explained, *“I tried to remove it myself... I didn’t think of tapping the alert and instead looked in the phone settings.”* When notifications disappeared without explicit interaction, participants formed incorrect assumptions or abandoned the task entirely. P7 assumed the threat had been automatically resolved, whereas P13 became confused when the alert vanished prematurely; lacking the knowledge to manually remove the threat, P13 chose to wait for help: *“The notification disappeared, so I couldn’t take any action... I would ask someone else for help.”* These findings demonstrate that alerts lacking clear interaction pathways cause misunderstandings and task abandonment.

A more critical issue was the lack of feedback on failures. In several instances, participants clicked “Resolve” to remove detected threats, but the system failed to execute the action and provided no subsequent notification of failure. Consequently, most users (P1, P6, P7) failed to recognize that the threat remained on the device, falsely assuming it had been resolved. In contrast, P10 noticed the suspicious application was still present but chose to cease further action, stating,

“Even though it feels like the threat is still there, I would probably stop... I don’t think it would cause serious harm since I didn’t really do anything in it.” These reactions indicate that users often treat their initial interaction (e.g., clicking a button) as task completion rather than actively verifying the outcome. Even when users recognize inconsistencies, the lack of guided remediation and a low perceived severity discourage further effort, leaving threats unresolved. Therefore, providing explicit feedback on both successful and failed actions is critical to supporting effective user responses.

Avoidance and Distortion. Even when users recognized a threat and understood the available actions, psychological factors shaped their responses. We identified three drivers. Fear of operation led to avoidance, insufficient motivation led to impulsive or delayed decision-making, and cognitive fatigue led to mindless compliance.

Fear of operation was heavily influenced by users’ technical backgrounds. Participants with computer science expertise (P5, P17) interpreted warnings more fluently, whereas non-experts—particularly older users—expressed fear and uncertainty. P2 described the interaction as intimidating: *“Even clicking on the warning feels scary. It’s like it’s alive... as if something might happen.”* Some participants avoided system interaction altogether; P12 attempted to locate and remove the application manually, while P13 disengaged entirely by powering off the device.

When systems failed to provide clear evidence or an explanation, they also failed to motivate users to act, and responses diverged into either impulsive action or delayed decision-making. Some participants acted quickly without fully understanding the situation (e.g., P18: *“I’ll probably delete it”*), whereas others postponed action until they could independently verify the risk. P15 selected “Skip” to seek additional information, stating, *“It didn’t feel that serious... I wanted to understand what exactly was risky... I can delete it whenever I want.”* When the necessity of an action is poorly justified, users appear to prioritize keeping control of their devices over following system recommendations.

Others followed system prompts without evaluating the threat at all. P22 stated, *“I don’t really know what it is, but I guess the app will take care of it... I don’t want to think about it for even a second,”* selecting the recommended action without further consideration. Cognitive burden can therefore suppress deliberate decision-making, leading users to bypass the motivation phase and comply without reflection.

Confusion and Uncertainty. Even when users actively attempted to respond, they remained confused about interface elements and unable to verify outcomes. Two drivers were at work: complex technical information and a lack of explicit feedback.

File paths displayed on warning screens confused non-technical users; P1, P15, and P18 mistook them for clickable URLs, while P8 repeatedly tapped the file path, assuming the interaction would trigger deletion. These misinterpretations show how developer-oriented information, presented without contextual cues, prompts erroneous user actions.

Even after completing a recommended action, participants were unsure whether the threat had been fully resolved. P10 noted, “*I don’t think just pressing the delete button solves everything,*” and P2 added, “*I wish it could assure me that my phone is back to the same state as before.*” The absence of post-action feedback left users anxious and, in some cases, more likely to engage in risky follow-up behaviors such as reinstalling compromised applications. Warning interfaces therefore need progressive information disclosure, offering clear, action-oriented guidance first and optional technical depth for advanced users, together with explicit confirmation once a threat has been removed.

These findings reveal that security warnings fail not at a single interaction point but across the entire response lifecycle. Even when users intend to act, unclear interaction design prevents successful execution; even when users understand the available actions, psychological factors distort their responses; and even after executing an action, the lack of feedback leaves them uncertain about the outcome.

Answer to RQ3: Ambiguous action labels such as **Resolve**, **Delete**, **Ignore**, and **Skip** induce conflicting mental models, leading users to overestimate their safety or hesitate out of anxiety. Technical details such as file paths further confuse non-expert users, and the absence of post-action feedback leaves them uncertain about whether threats persist. Effective mobile warnings must employ outcome-oriented language, minimize extraneous technical information, and confirm successful remediation.

5 Discussion

By mapping observed failures to the C-HIP model, we show that ineffective mobile security is rarely a simple visual flaw; it is a cascading breakdown across the human information processing pipeline, from initial attention to behavioral motivation. Below we discuss the cognitive gaps behind these breakdowns, the interface properties that produce them, how far our findings are likely to generalize, and what they imply for warning design.

Mismatched Mental Models and Security Misconceptions. Participants frequently exhibited limited or inaccurate understandings of malware threats, often relying on incomplete or incorrect mental models. Rather than recognizing malicious intent, many users characterized malware as benign software bugs or routine development errors (*e.g.*, P16, P22). This confusion between functional flaws and malicious exploits reflects a deeper breakdown in conceptual understanding. Prior work has shown that technical terminology in security warnings often fails to resonate with non-expert users, impairing both comprehension and decision-making [9,24]. These misconceptions were further shaped by contextual cues; participants often dismissed warnings from sources they trusted (*e.g.*, a

familiar bank) or found the message content too appealing to ignore. Such patterns align with prior findings that users downgrade warning severity based on perceived legitimacy or familiarity [20,35]. Moreover, while habituation to frequent alerts is a well-documented mechanism in prior warning research [3,6,14], participants in our study specifically reinterpreted legitimate antivirus warnings as advertisements or routine system notifications, which markedly reduced the likelihood of protective action. To our knowledge, this misattribution pattern is a novel empirical contribution.

Usability Barriers in Warning Interfaces. Even when antivirus warnings were noticed, interface design often obstructed effective responses. Participants regularly misunderstood action labels such as “Resolve,” “Delete,” or “Ignore,” fostering dangerous misconceptions about whether a threat was fully addressed or merely silenced.

Detection mechanisms further dictated these outcomes. While automatic scanning facilitates timely mitigation, manual scanning creates security gaps by misaligning with mobile notification behaviors. Unlike desktop environments, where alerts often receive immediate attention, mobile security prompts compete for attention within secondary layers such as notification drawers. Our observations suggest that these mobile-specific constraints—in which security cues are “hidden by design”—place the burden of surfacing critical threats on the user. The high failure rate in manual scanning appears consistent with the passive security interaction noted in prior work, in which users accustomed to background-operating “plugin” security tools may struggle to exercise initiative in response to active threats [25]. Participants’ frustration with the manual effort required to locate scan notifications supports this reading, as P19’s remark on the poor visibility of preventive tools illustrates. Such uncertainty undermines trust and increases the likelihood of risky follow-up behaviors, such as reinstalling compromised applications when system feedback remains ambiguous.

Generalizability Across Mobile Threat Vectors and Cultural Contexts. Our study is grounded in a single attack vector, SMS-based phishing leading to a trojanized APK installation, so other mobile threat vectors such as QR codes and fake app stores were not directly evaluated. To address generalizability, we categorize our findings into two levels: interface design failures, which are usability and interaction breakdowns stemming from the structural and visual properties of the user interface, and context-driven interpretive failures, which depend on how users interpret alerts within a given scenario.

Interface design failures, such as the low accessibility of manual-scan prompts, the ambiguity of action labels, and the lack of behavioral guidelines, stem from the properties of the warning interface itself, irrespective of the path through which the warning was triggered. These failures are therefore likely to generalize across mobile malware distribution scenarios. Our explanation for the high failure rate of manual scans, that users accustomed to background-operating security apps engage passively with threats, rests on existing research regarding South

Korea’s mandatory software ecosystem [25] rather than on evidence collected here. The low discoverability of manual-scan prompts is itself a structural property likely to generalize, but whether users in other regulatory contexts show the same degree of passive engagement remains an open question for cross-national comparative studies.

Context-driven interpretive failures are more tightly bound to the scenario. P18 read the “**Resolve**” label as a remedy provided by the bank, and P14 re-framed the alert as a conflict between the bank’s security level and the phone, both shaped by cues specific to SMS-based phishing such as the perceived legitimacy of the impersonated institution. Different threat vectors place users in distinct narrative contexts and may therefore produce different scenario-specific misunderstandings even when the ambiguous label is the same.

Design Implications for Effective Mobile Antivirus Warnings. Our results suggest three evidence-based directions to realign mobile antivirus warnings with users’ mental models and the C-HIP framework. The first involves refining terminology and layout to support the *comprehension* and *beliefs* stages. Labels such as “**Resolve**” or “**Ignore**,” while technically accurate to developers, were interpreted inconsistently by casual users; P6 read “**Resolve**” as partial remediation, and P13 assumed “**Ignore**” blocked execution. Replacing these with precise, action-oriented alternatives such as “**Remove malware now**” aligns the system’s intent with user perceptions. To further reduce decision friction, the secure path should be presented as a prominent, single primary button, since simplifying the choice architecture minimizes the visual weight of technical metadata and risk-accepting alternatives.

A second recommendation is to default to automatic threat detection so that the information processing pipeline is initiated without user intervention. As illustrated by Kaspersky, manual scanning placed excessive responsibility on users and often failed at the *attention* stage, whereas automatic scanning in Avast and V3 prompted more timely responses. The structure of user options also influences the transition from *motivation* to *behavior*. V3, which offered a single “**Delete**” button, achieved the highest removal rate, whereas Avast’s binary choice between “**Ignore**” and “**Resolve**” introduced uncertainty and decision friction, suggesting that reducing optionality when a secure path is strongly recommended can promote safer outcomes. We suggest calibrating this optionality to detection confidence. When confidence is high, provide a single action such as “**Delete**”; when confidence is low, quarantine the file and offer “**Delete**” and “**Restore**” so that the user can reconsider once the file is safely isolated.

Lastly, antivirus interfaces must confirm the outcome of user actions, reinforcing safe *attitudes* for future interactions. Many users remained anxious after selecting “**Delete**” or “**Resolve**,” unsure whether the threat had been fully removed. Systems should close this gap with explicit indicators such as “**Suspicious file successfully deleted**” or “**No further threats detected**,” presented as full-screen confirmations rather than transient toast notifications.

Effective warnings thus require more than accurate detection; they require an interface that aligns with users’ mental models and decision-making styles.

6 Limitations

Our study has several limitations that should be acknowledged when interpreting these findings.

Sample and Scope. While we carefully recruited participants to reflect typical South Korean smartphone users, our findings are limited to a single cultural context and the Android platform. Although we deliberately selected three antivirus applications with distinct interfaces and detection approaches, results might differ across other cultures, iOS devices, and additional products, as users from different regions may possess varying mental models and expectations. Furthermore, while the relatively small sample size limits the generalizability of our findings, our focus on South Korea offers valuable insights into mobile security behaviors in an advanced digital environment. South Korea’s 96% smartphone penetration and 4.51 hours of daily use, ranking 9th globally, exceed global averages and represent a highly sophisticated threat landscape [23,39]. Our recruitment also did not screen for prior security training; we instead characterized participants’ security background after the task through the background questions and knowledge quiz. The resulting sample showed only moderate security familiarity, but we cannot rule out that unmeasured prior exposure shaped some participants’ responses.

Ecological Validity. For ethical reasons, our laboratory study cannot fully replicate the complexity of real-world mobile contexts where users face competing demands for attention. To enhance ecological validity, we built high-fidelity artifacts impersonating a real banking institution. While laboratory constraints remain, this controlled approach enabled a systematic comparison across antivirus interfaces and captured genuine user responses while ensuring participant safety through study-provided devices. Our experiments were also conducted on Google Pixel 3a smartphones (5.6-inch displays) with the application versions available at the time of data collection, and both the interfaces and the platform have since evolved. The structural properties we identify, such as the discoverability of scan prompts and the ambiguity of action labels, are unlikely to be tied to specific releases, but the precise wording and layout of individual warnings may differ in existing versions. Future work should examine how the larger screens of contemporary devices and evolving Android notification policies may influence visual attention and threat awareness.

Study Design Trade-offs. This study utilized a within-subjects design where participants engaged with three commercial antivirus applications in a single,

fixed mobile malware phishing scenario. This choice reflects a prevalent and realistic mobile threat vector observed in practice and allowed us to isolate the effects of interface design. A between-subjects design would have better approximated a first encounter with an unfamiliar threat, but it would have confounded interface differences with individual variation in security knowledge and risk attitude, and would have required a far larger sample than is feasible for interview-based analysis. To mitigate learning effects, we counterbalanced the order of applications using a Latin square design. Although the exclusion of one participant resulted in a final sample of $N = 23$, which prevented perfect counterbalancing, this slight imbalance does not compromise our primary findings: our core contribution derives from qualitative analysis of user behavioral patterns and the root causes of misconceptions rather than from precise quantitative measurement, so the deviation did not affect the theoretical saturation of the qualitative themes. Examining mitigation success by trial position, pooled across the three applications, we observe a modest upward trend of 47.8% (1st trial), 52.2% (2nd trial), and 56.5% (3rd trial), suggesting that repeated exposure to a similar attack allowed for some adaptation over time. Our estimates of user misunderstanding are therefore likely conservative; in real-world settings where attacks occur in isolation, we expect user confusion to be more pronounced. Importantly, when restricting the analysis to each participant’s first exposure only, the relative ordering of applications (V3: 75.0% (6/8) > Avast: 37.5% (3/8) > Kaspersky: 28.6% (2/7)) matches the ordering observed in the full pooled results reported in Section 4.2, suggesting that our core interface-design finding is unlikely to be purely an artifact of trial order.

Data Collection Methods. Our qualitative analysis combined behavioral observations with post-task interviews, using the actual warning screens encountered to enhance recall accuracy. While self-reported explanations may be subject to social desirability bias or incomplete recollection, the triangulation of observed behaviors with participant explanations provided rich insights into both actions and underlying mental models. Additionally, our focus on immediate responses limits our understanding of how security behaviors might evolve with repeated exposure or long-term antivirus use.

7 Conclusion

Our study characterizes usability barriers in mobile antivirus warnings that impede security outcomes. Through a task-based analysis of three widely used applications in South Korea, we show that existing interface designs fail to bridge the gap between technical detection and effective user response. Detection modality largely dictates mitigation success: automatic scanning facilitates timely resolution, whereas user-initiated workflows frequently fail at the attention stage. These gaps are widened by ambiguous action labels such as “Resolve,” “Ignore,” and “Skip,” and by insufficient system feedback, which together foster uncertainty and risky follow-up behaviors.

Our results underscore that accurate detection is necessary but not sufficient for protection. Effective mobile security requires user-centered designs that clearly communicate consequences, guide remediation, and sustain trust. Future research should prioritize post-remediation feedback, interfaces tailored to varying levels of expertise, and cross-cultural validation of usability challenges across diverse regulatory environments.

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Appendix

Interview Questionnaire

Part 1. Antivirus Software Usage Experience.

Q1: Have you ever heard of “Antivirus software” or “Vaccine software”? If so, please explain the purpose of antivirus software. [If participants are unfamiliar with both terms, we provide the names of well-known antivirus products in the Republic of Korea (*e.g.*, ‘ALYac’³ and ‘V3’⁴) to proceed with the interview.]

Q2: Antivirus software is a security program designed to prevent, detect, search for, and remove various types of malware from computers, networks, and other devices. Have you ever used antivirus software on your smartphone? If yes, please share the list of antivirus software you currently use (or have used) with us.

Q2-1: When, how often, and which features of the antivirus software do (or did) you primarily use? [if applicable]

Q3: What reasons led you to (start using it / start and then stop using it / decide not to use it)?

Part 2. Risk Perception and Responses to Antivirus Warnings

Imagine you receive the following text message while looking for a loan due to financial difficulties. Using the provided experimental smartphone, freely perform the actions you would typically take in this real-world situation. If possible, please verbalize your thoughts as you go through the process.

³ ‘ALYac’ developed by EST Security (<https://en.estsecurity.com/product/alyac>)

⁴ ‘V3’ developed by AhnLab (<https://www.ahnlab.com/en/product/v3-mobile-plus>)

[IBK Corporate Bank] Special Support Approval Notification
 Hello, valued customer,
 You are eligible for a government-supported special guarantee loan program.
 However, your application needs to be confirmed. Please complete the application by 16:00 today by following these steps:

1. Click the application link provided below.
2. Click “Mobile Application” and then download the application.
3. Allow installation from unknown sources to complete installation.
4. Enter your information and click the “Apply” button.

Your secure application link: <https://bit.ly/XXXXXX>

Q1: Why did you or didn’t you click on the URL in the text? [Only for the first iteration.]

Q2.1: Why did you stop the installation? [If the participant deleted the IBK app]

- Which warning element had the greatest influence on your decision to stop the installation?
- Did you experience any difficulties or concerns when deleting the app?
- Did you consider reporting the IBK app?
- Do you think the app was deleted?

Q2.2: Why did you stop the installation? [If the participant installed the IBK app]

- What actions do you take afterward?

Q3: What did you think when you saw the security warning? Please explain the warning as you understand it. If you did not understand it, please explain why it was difficult to understand.

Q4: What do you think could happen if you ignore the warning?

[After completing three iterations]

Q5: (For Kaspersky and V3 Mobile Security only) Please navigate to the path indicated in the warning and manually remove the malware from the experimental smartphone.

Q6: Please rank the three antivirus apps based on which was the most helpful. What are the reasons for your ranking?

Q7: After removing the malware, are there any other actions you think the antivirus should take?

Q8: Do you use this bank as your primary bank? Did that influence how you responded to the warning?

Part 3. Demographic and Security Knowledge.

Q1: What gender do you identify with?

Q2: What is your age range?

- 19-28
- 29-38
- 39-48
- 49-58
- 59+

Q3: What is the highest degree or level of school that you have completed?

- Middle school or Less
- High school or equivalent
- Associate's Degree
- Bachelor's degree
- Master's degree
- PhD
- Others
- I prefer not to answer

Q4: Do you have any experience in computer science? If so, what is it?

Q5: Which of the following is not a type of malware?

- Worm
- Trojan
- Firewall
- Virus
- I don't know

Q6: What is the main purpose of ransomware?

- It places unwanted advertisements on your computer.
- It monitors and tracks the user's search history.
- It encrypts the user's files and asks for money to restore them.
- It takes control of your computer and deletes files.
- I don't know

Q7: How does the 'Drive-by Download' method distribute malware?

- Malware is inserted into websites visited by users and automatically downloaded.
- It is distributed by encouraging users who receive emails containing malicious code to download attached files or click links.
- It spreads malware using Wi-Fi networks connected to users.
- Files containing malware are spread through file-sharing programs.
- I don't know

Notices. This study has no relation to any of the companies or products (*e.g.*, IBK Corporate Bank, Avast, Kaspersky, or V3 Mobile Security) mentioned during the interview process. We clearly state that the SMS text message, mobile website, and banking app used during this interview process were created by the researchers as a fictitious scenario for the study.

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